



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (Civil) (CBCS) Degree Examinations
December 2023

Duration: 3 Hours

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

20 Marks

- The Fundamental equation of fluid statics is**
 - Bernoulli's equation
 - Euler's equation
 - Pascal's equation
 - Archimedes' principle
- The property of fluid that determines resistance to flow**
 - Density
 - Viscosity
 - Pressure
 - Temperature
- Type of flow that occurs when the Reynolds number is less than 2000**
 - Laminar flow
 - Turbulent flow
 - Steady flow
 - Compressible flow
- Device used to measure the velocity of flow in a fluid**
 - Venturimeter
 - Manometer
 - Pitot tube
 - Orifice meter
- Equation that describes the conservation of mass for an incompressible fluid in a steady flow**
 - Bernoulli's equation
 - Poisson's equation
 - Continuity equation
 - Euler's equation
- Type of head loss that occurs when the flow velocity increases due to a decrease in pipe diameter**
 - Frictional head loss
 - Minor head loss
 - Major head loss
 - Turbulent head loss
- In pipe flow, the Darcy – Weisbach equation is commonly used to calculate which of the following parameters**
 - Flow rate
 - Flow velocity
 - Pressure drop
 - Pipe diameter
- The primary purpose of an orifice in fluid mechanics**
 - To measure flow rate
 - To increase flow velocity
 - To control fluid temperature
 - To reduce fluid pressure
- The parameter commonly used to classify open channel flow regimes**
 - Reynold's number
 - Froude number
 - Mach number
 - Strouhal number
- The type of flow that occurs when Froude number is equal to 1 in open channel flow**
 - Critical flow
 - Supercritical flow
 - Subcritical flow
 - Laminar flow
- Mercury does not wet glass due to its property of**
 - Cohesion
 - Adhesion
 - Surface tension
 - viscosity
- Term for the ratio of actual discharge to theoretical discharge for an orifice**
 - Coefficient of discharge
 - Coefficient of velocity
 - Coefficient of contraction
 - Coefficient of resistance
- The primary principle involved in the context of impact of jet on vanes**
 - Pascal's principle
 - Newton's third law
 - Archimedes' principle
 - Bernoulli's principle

14. In the context of impact of jet on vanes, what is meant by "momentum transfer"
 A) Transfer of pressure
 B) Transfer of fluid particles from jet to vane
 C) Transfer of angular momentum
 D) Transfer of kinetic energy from jet to vane
15. The height of capillary tube to which a liquid will rise or fall depends on
 A) The width of the tube
 B) The temperature of the liquid
 C) The density of the liquid
 D) The Combination of liquid properties and tube dimensions
16. The Parameter which is typically used to measure the flow rate of a fluid through a notch
 A) Pressure
 B) Velocity
 C) Depth of flow
 D) Temperature
17. The type of notch that is typically used for measuring low flow rates
 A) Rectangular notch
 B) Triangular notch
 C) Trapezoidal notch
 D) Venturi notch
18. Turbines are typically used to
 A) Pump fluids
 B) Generate electricity from fluid flow
 C) Measure fluid viscosity
 D) Increase fluid pressure
19. Type of pump that is commonly used for high flow, low pressure applications and is often used in heating and cooling systems
 A) Centrifugal pump
 B) Positive displacement pump
 C) Diaphragm pump
 D) Gear pump
20. The unit of measurement for friction loss in fluid flow
 A) Meters per second (m/s)
 B) Newton's(N)
 C) Pascal's (Pa)
 D) Meters(m)

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the distinction between solids and fluids.	04	L*1	1	1
	b) Calculate the density, specific weight and weight of one liter of petrol of specific gravity = 0.7.	04	L2	1	1
	c) State and prove Pascal's law.	08	L2	2	2
2.	a) Explain: i) Steady flow and unsteady flow. ii) Uniform and non-uniform flow.	04	L2	2	1
	b) Differentiate between absolute pressure and gauge pressure.	04	L2	2	2
	c) A simple U-tube manometer containing mercury is connected to a pipe in which a fluid of sp. gr. 0.8 and having vacuum pressure is flowing. The other end of the manometer is open to atmosphere. Find the vacuum pressure in pipe, if the difference of mercury level in the two limbs is 40 cm and the height of fluid in the left from the center of pipe is 15 cm below.	08	L2	1	2
3.	a) Differentiate between a simple manometer and a differential manometer. What are their respective applications?	04	L2	2	1
	b) What do you understand by 'Total Pressure' and 'Centre of Pressure'?	04	L2	2	2
	c) Assess the impact of surface tension on capillarity and provide examples for these phenomena.	08	L3	1	2
Unit – II					
4.	a) What is the primary purpose of a Pitot tube in fluid mechanics?	04	L2	3	1
	b) Water enters a horizontal pipe of non-uniform cross-section with a velocity of 0.6m/s and leaves the other end with a velocity of 0.4m/s. At the first end, pressure of water is 1600N/m ² . Calculate the pressure of water at the other end. Density of water = 1000 kg/m ³ ?	04	L1	3	1
	c) Derive the equation for the flow rate (discharge) over a V-notch or triangular-notch.	08	L2	4	2

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| 5. | a) | Explain water hammer is in the context of fluid dynamics within pipes | 04 | L2 | 3 | 2 |
| | b) | State Bernoulli's energy equations with assumptions and limitations. | 04 | L1 | 3 | 1 |
| | c) | A rectangular notch 0.5m wide has constant head of 400 mm. Find the discharge over the notch in liters per second, if the coefficient of discharge for the notch is 0.62. | 08 | L2 | 4 | 2 |
| 6. | a) | Explain the hydraulic coefficients of an orifice and the relationship between them. | 04 | L1 | 4 | 2 |
| | b) | Provide the equations for the discharge over rectangular notch. | 04 | L2 | 4 | 2 |
| | c) | A venturi-meter of 15 cm inlet diameter and 10 cm throat is laid horizontally in a pipe to measure the flow of oil of 0.9 specific gravity. The reading of a mercury manometer is 20 cm. Calculate the discharge in lit/min. | 08 | L2 | 3 | 1 |

Unit – III

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| 7. | a) | Describe the concept of work done and efficiency in the context of the impact of jet on vanes. | 04 | L2 | 5 | 2 |
| | b) | Define open channel flow. Highlight the key differences between pipe flow and open channel flow. | 04 | L2 | 5 | 2 |
| | c) | Calculate the dimensions of the rectangular cross-section of an open channel which requires minimum area to convey 10 m ³ /s. The slope being in 1500. Take the Manning's N as 0.013. | 08 | L2 | 5 | 2 |
| 8. | a) | Discuss the impulse-momentum equation and its application in the context of the impact of jet on vanes. | 04 | L1 | 5 | 1 |
| | b) | Explain how the force is exerted by a jet on fixed and moving vanes. | 04 | L1 | 5 | 1 |
| | c) | Explain the concept of most economical channel sections and derive the conditions for the most economical trapezoidal channel sections. | 08 | L2 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
